

B4M36SAN
BE4M36SAN
Seminar/Tutorial

Alikhan Anuarbekov

Ali

anuarali@fel.cvut.cz

- **You have 4 weeks left to form a team (deadline 11.11.2024)**
 - 4 people maximum
 - We strongly recommend you start forming and preparing for Final assignment
- **After deadline you won't have enough time to discuss your topic!**
 - **1 week** to come up with idea, **2 week** to prepare a detailed plan

https://cw.fel.cvut.cz/wiki/courses/b4m36san/new_final_assignment

Steps

1. Team forming.
2. A few sentences proposing the work topic: we will provide feedback to calibrate its difficulty before you start working on a plan.
(submitted as txt in BRUTE)
3. You will hand in your plan (D1) at least a day before a consultation. Consultation will be per team with a tutor. We will specify some timeslots, and teams will sign up individually.
4. You will hand in your report (D2) one week before the review deadline.
5. You will hand in your review (D3) a few days before the presentation.
6. You will present your findings and then submit the presented slides and poster (D4).

Edit

Steps in Time

Task	Date/Deadline
1.	finished at the lab in week starting 11.11.
2.	17.11. (two weeks before consultation)
3.	24.11. (consultations follow)
4.	1.1. (two weeks before 5.)

Alternative: I assume that I know the TYPE (model) of formula between Y and X1/X2/....
Then I can derive relations for every human independently

For example: Human age = V is increasing -----> this causes the H=height to grow
Or other way around: H=height is increasing ----> this means that human ages

Assume: $H = f(V)$ or $f(V, H) = 0$ most easiest function is linear:

$$H = \beta_0 + \beta_1 \cdot V \quad \text{or} \quad \alpha_0 + \alpha_1 \cdot H + \alpha_2 \cdot V = 0 \quad \text{Linearity assumption}$$

But there are other factors, reality isn't that primitive:

- * How often does this human eats a day? Note: discrete(categorical) variable!
- * Does he/she have some disease that makes it age faster? Note: this too = Yes/No

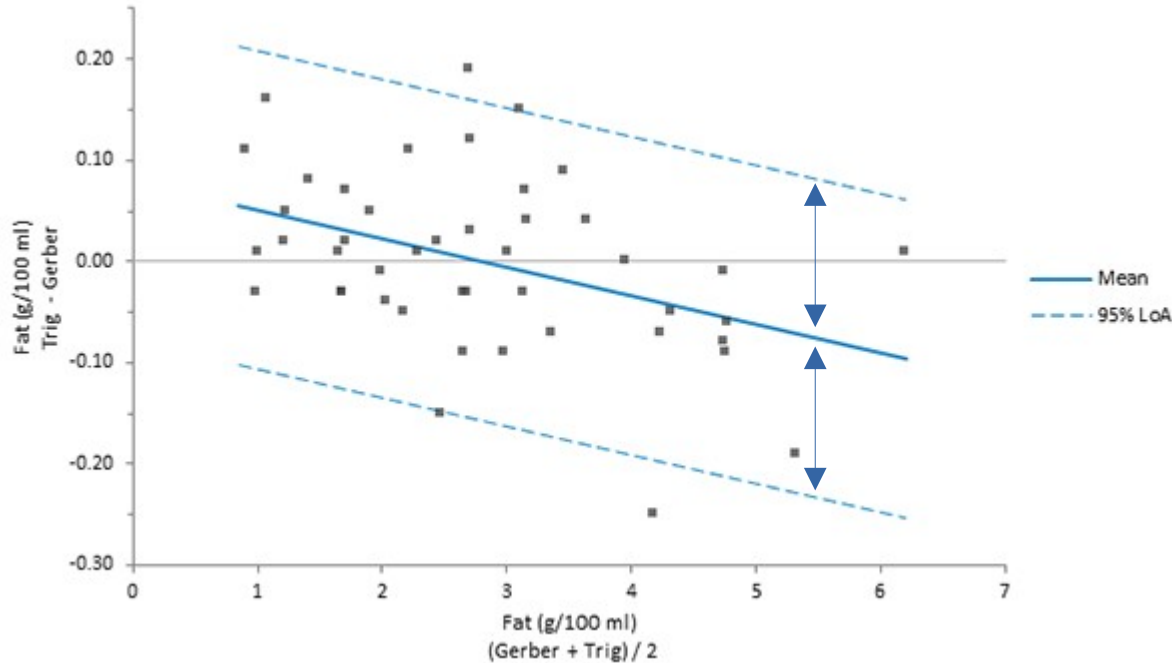
Lets just say that all the other factors are secondary and approximate them:
(Central Limit Theorem)

$$hidden_1 + hidden_2 + \dots \approx \epsilon, \quad \epsilon \in N(\underline{0}, \sigma^2) \text{ (if mean is not zero, just move it to intercept)}$$

$$H = \beta_0 + \beta_1 \cdot V + \epsilon, \quad \epsilon \in N(0, \sigma^2) \quad \text{Normality of error}$$

$$H = \beta_0 + \beta_1 \cdot V + \epsilon, \quad \epsilon \in N(0, \sigma^2)$$

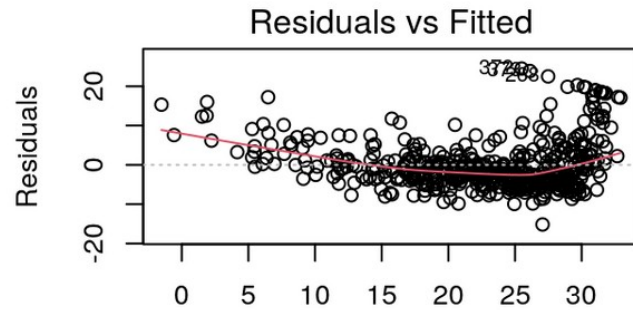
We say that secondary factors are limited by some effect power = σ



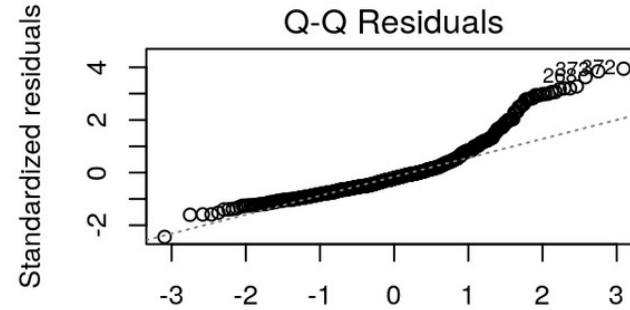
Since these effects shouldn't change the slope (vertical tilt / incline $\updownarrow$), it should have same variance across entire axis = **Homoscedasticity assumption**

<https://library.virginia.edu/data/articles/diagnostic-plots>

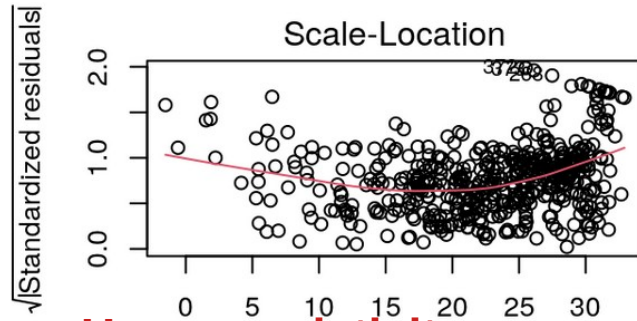
Seminar 2: LREG_boston_intro.html



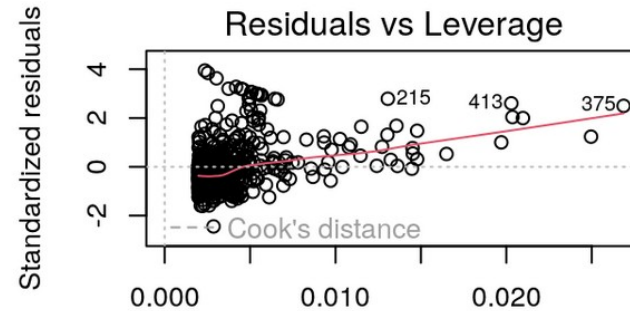
Linearity



Normality



Homoscedsticity

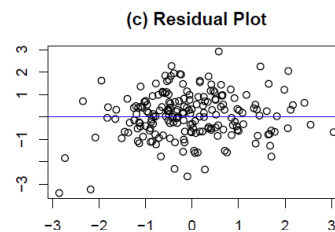
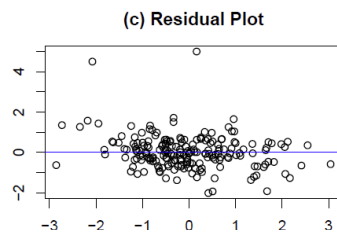
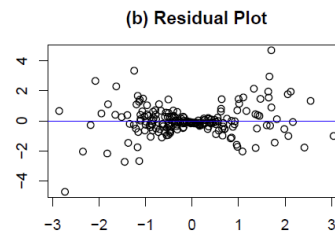
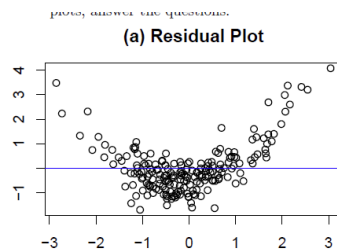
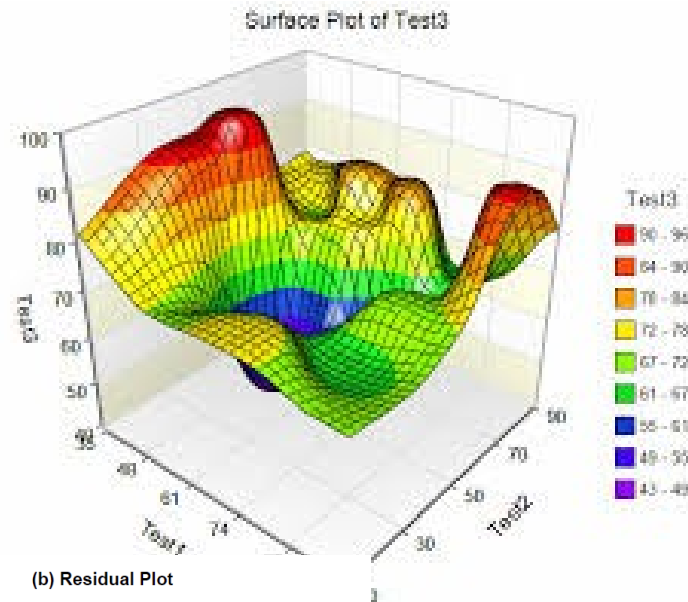
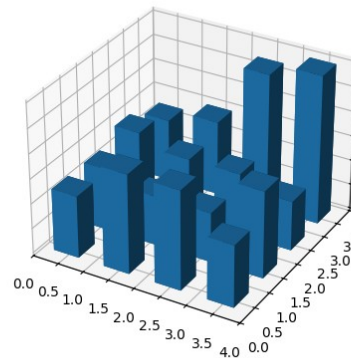
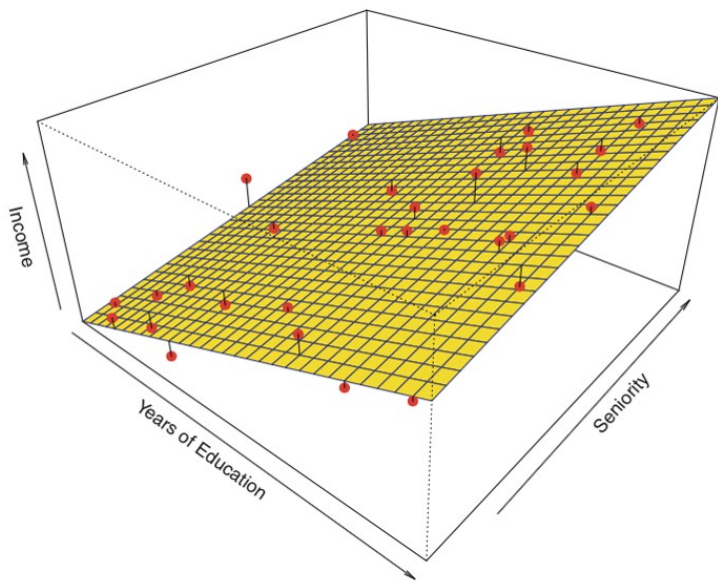


Influential points/Outliers?

$$\text{Income} = \beta_0 + \beta_1 \cdot \text{Years} + \beta_2 \cdot \text{Seniority} + \epsilon,$$

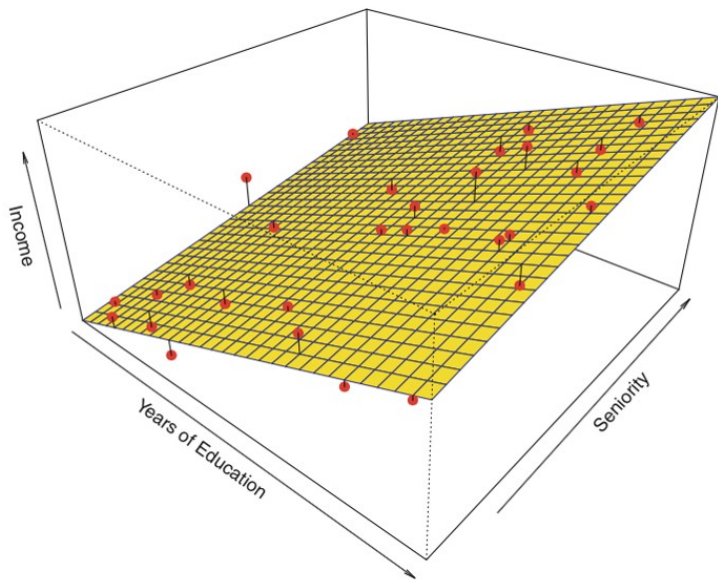
$\epsilon \in N(0, \sigma^2)$ Variance now is an surface in 3D, not just

More than 1 independent variable:
Normality? Homoscedasticity?



$$\text{Income} = \beta_0 + \beta_1 \cdot \text{Years} + \beta_2 \cdot \text{Seniority} + \epsilon, \quad \epsilon \in N(0, \sigma^2)$$

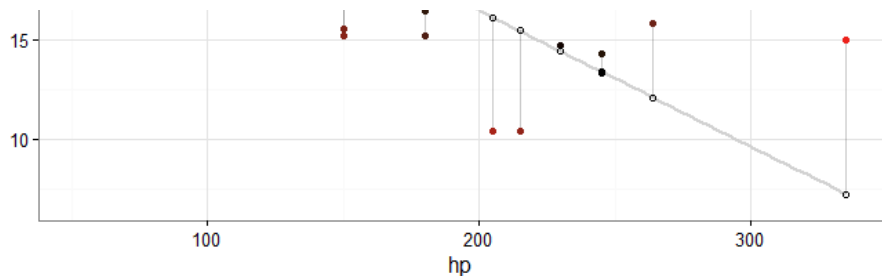
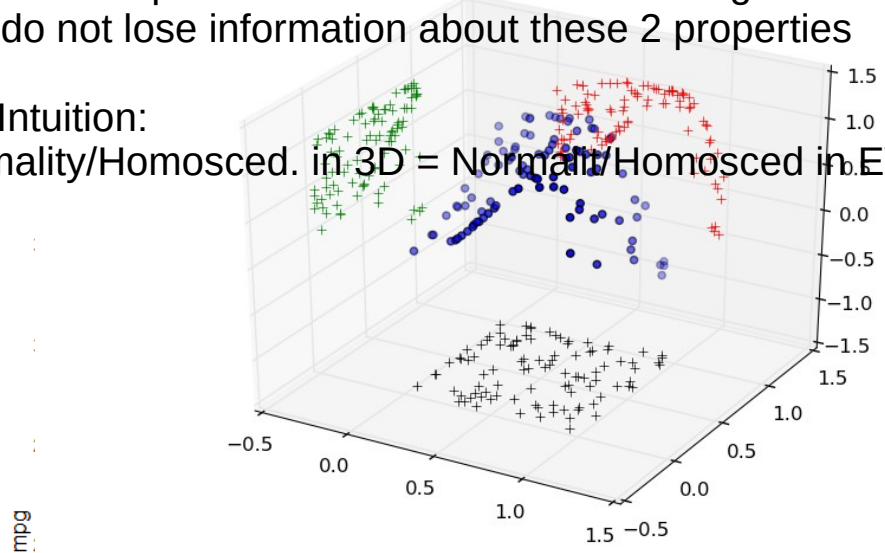
More than 1 independent variable:
Normality? Homoscedasticity?



- > Normality/Homosced. does not depend on surface shape, only on deviations from this surface in Y axis = residuals
 - * You can see them as lines on the left image
- > If I were to replace all X variable axes with single residual axis, I do not lose information about these 2 properties

Why? Intuition:

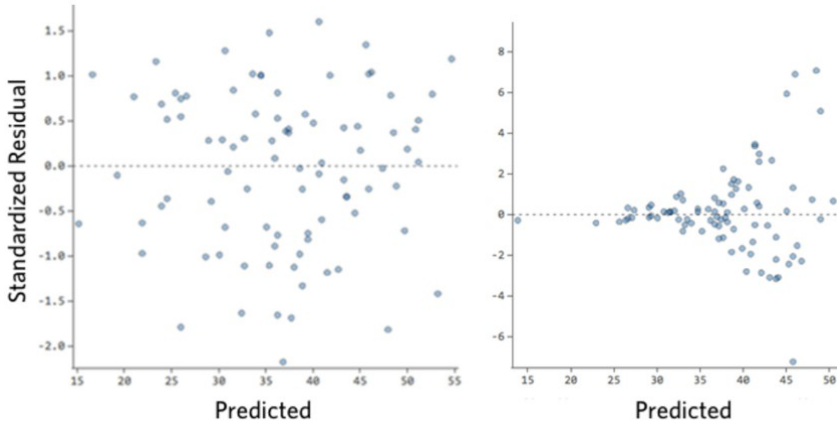
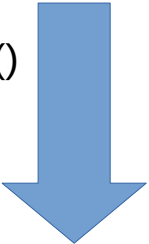
- > Normality/Homosced. in 3D = Normality/Homosced. in EVERY AXIS X



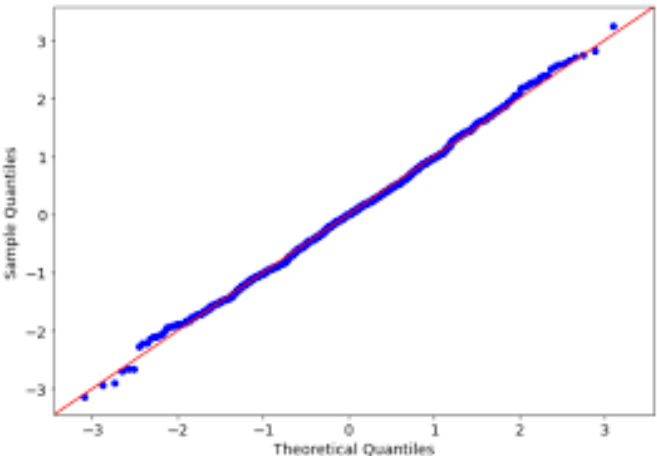
Okay, I see homoscedasticity, but normality?

More formal way of normality check – QQ plot:

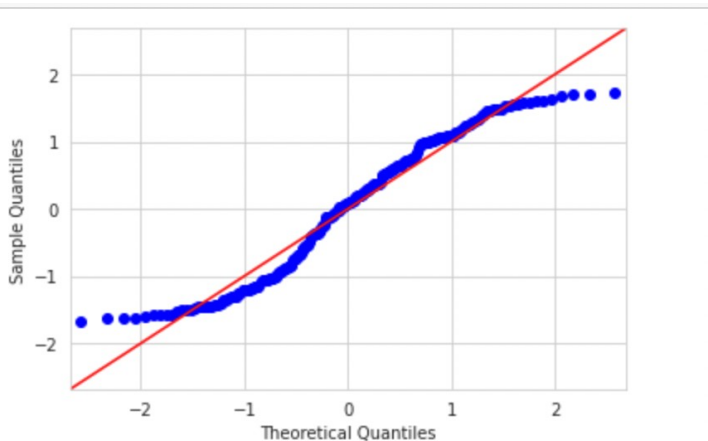
Call QQPlot()



Surely is normally distributed



Not quite normal



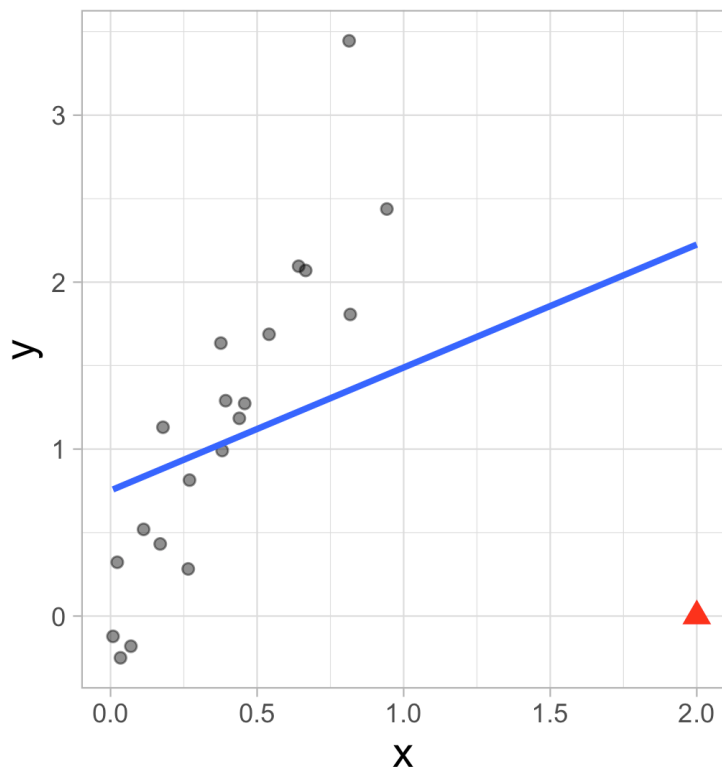
influential point

> Outlier or point on right side (large X value)

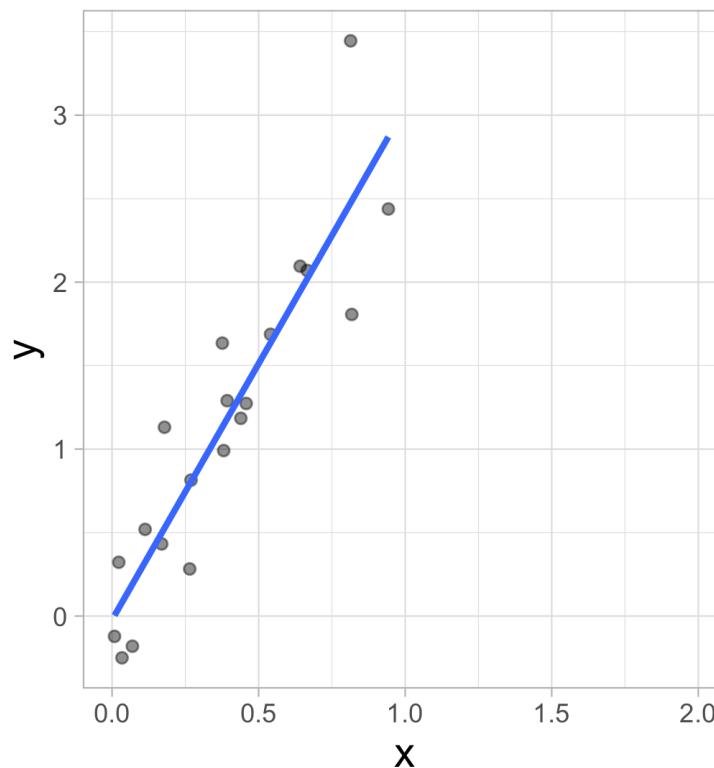
For every point: compare slope with point and without point.

If it is larger than a given threshold, we mark this point as **influential point**

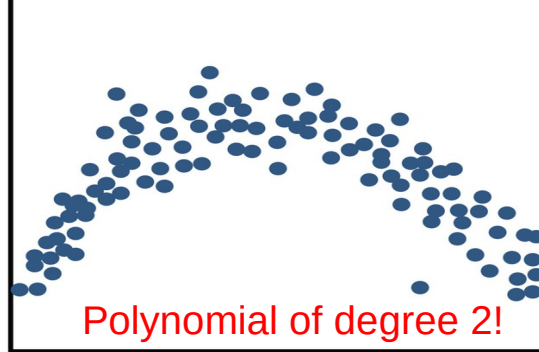
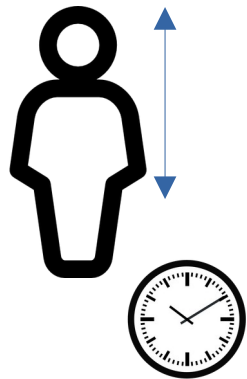
With Influential Point



Without Influential Point



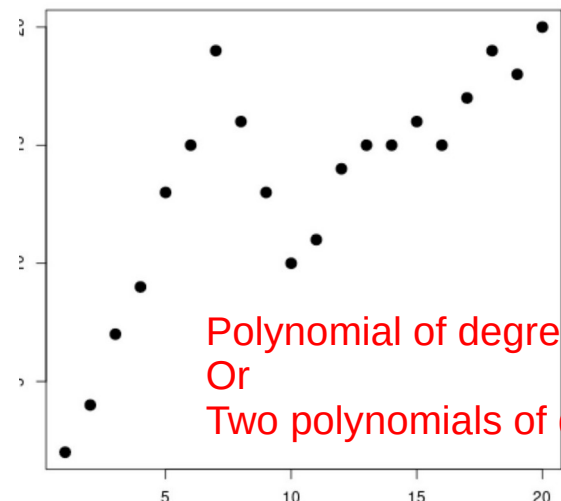
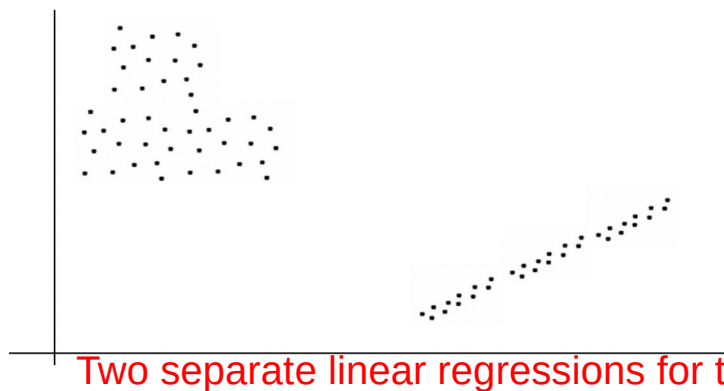
Now, again to 2 variables:



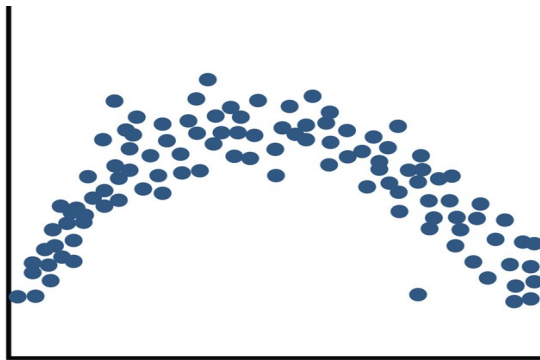
In all cases:

- > Homoscedasticity is broken
- > Normality (mean=0) is broken
- > Data are not linear
- > previous model will surely fail
- > But I know math of each of this shape!

Y = Height, X = Age
Both are random variables

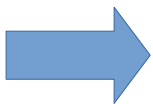


If I see something like:



From this:

$$Y = \beta_0 + \beta_1 \cdot X + \epsilon, \\ \epsilon \in N(0, \sigma^2)$$



How do I fit it? Use multivariate approach:

> creating synthetic data from given data

$$\begin{aligned} X_1 &= X \\ X_2 &= X^2 \\ &\dots \\ X_p &= X^p \end{aligned}$$

Then I want a Polynomial of degree 2

It will look like this:

$$Y = \beta_0 + \beta_1 \cdot X + \beta_2 \cdot X^2 + \epsilon, \\ \epsilon \in N(0, \sigma^2)$$

$$Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \dots + \beta_p \cdot X_p + \epsilon, \\ \epsilon \in N(0, \sigma^2)$$

Polynomial regression

Some strange R poly() stuff? Well, not really:

$$\begin{array}{l} X_1 = X \\ X_2 = X^2 \\ \dots \\ X_p = X^p \end{array} \quad \vec{X} = \begin{pmatrix} 1 \\ 2 \\ \dots \\ 100 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1^2 & 1^3 \\ 2 & 2^2 & 2^3 \\ \dots & \dots & \dots \\ 100 & 100^2 & 100^3 \end{pmatrix}$$

But they are correlated!
May cause some problem, since:

correlation = broken assumption of LM

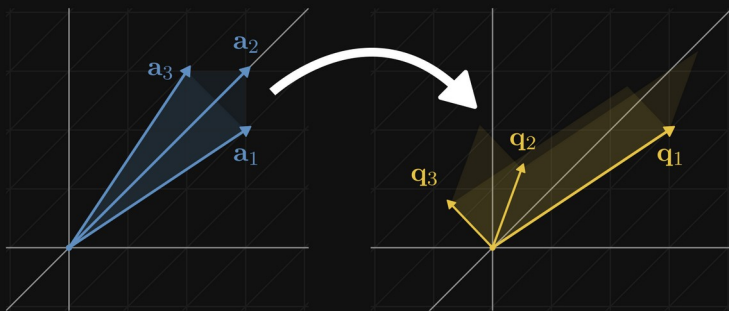
$$\vec{X} = \begin{pmatrix} 1 \\ 2 \\ \dots \\ 100 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1^2 & 1^3 \\ 2 & 2^2 & 2^3 \\ \dots & \dots & \dots \\ 100 & 100^2 & 100^3 \end{pmatrix} \rightarrow \text{Gramm-Schmidt ortog} \rightarrow \begin{pmatrix} -0.17 & 0.22 & -0.25 \\ \dots & \dots & \dots \\ \dots & \dots & \dots \\ 0.15 & 0.14 & 0.21 \end{pmatrix} = \vec{X}_{\text{ortog}}$$

Even though they are changed = changed coefficients,

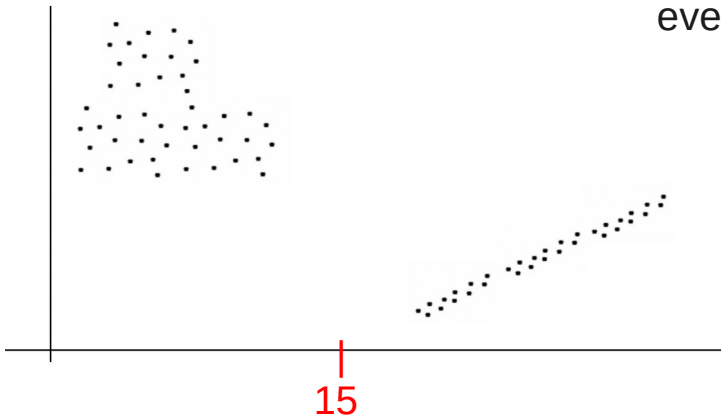
but their predictions are the same!

`predict(lm with x coefs) = predict(lm with x_ortog coefs)`

The Gram-Schmidt process



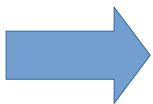
If I see something like:



Then I want two separate linear models on every interval

From this:

$$Y = \beta_0 + \beta_1 \cdot X + \epsilon, \\ \epsilon \in N(0, \sigma^2)$$



It will look like this:

$$Y = \beta_0 + \beta_1 \cdot X + \epsilon, \text{ if } X < 15 \\ Y = \beta_2 + \beta_3 \cdot X + \epsilon, \text{ if } X \geq 15 \\ \epsilon \in N(0, \sigma^2)$$

How do I fit it? Train each interval separately, e.g., divide data into 2 separate LM calls

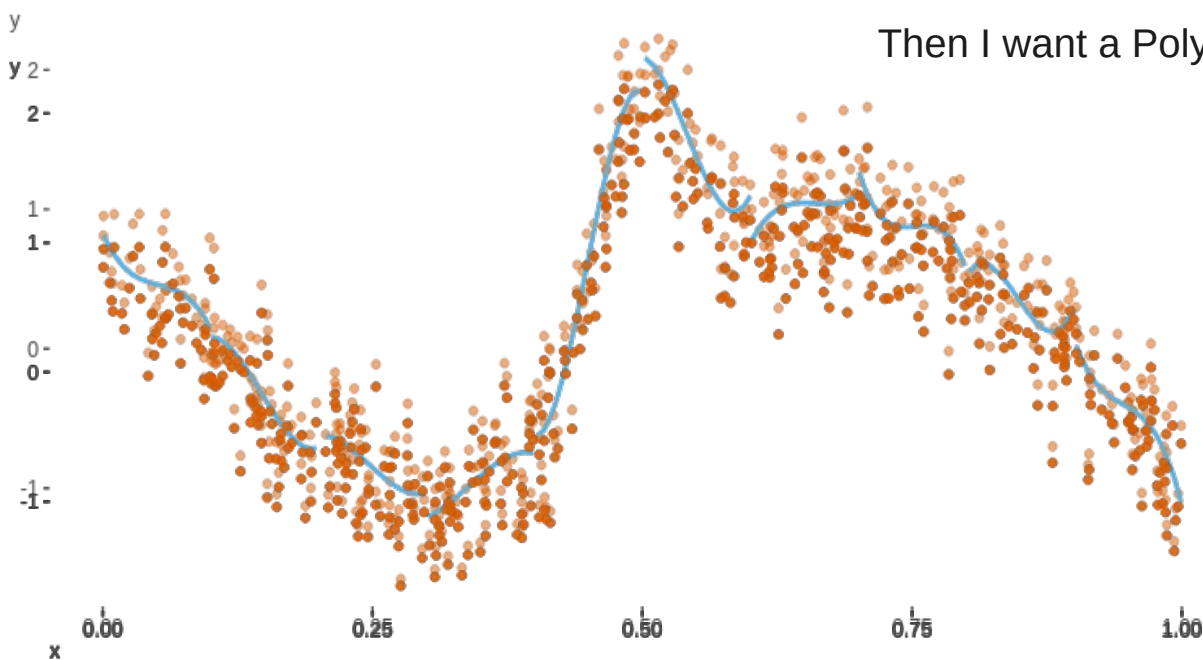
OR

Add a categorical switch variable for every interval change = knot

$$Y = \beta_0 + \beta_1 \cdot X + \beta_2 \cdot (X \geq 15) + \epsilon \\ \epsilon \in N(0, \sigma^2)$$

Step functions

If I see something like:



Then I want a Polynomial of large degree

$$X_1 = X$$

$$X_2 = X^2$$

...

$$X_P = X^P$$

$$Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \dots + \beta_P \cdot X_P + \epsilon,$$

$$\epsilon \in N(0, \sigma^2)$$

Piece-wise polynomial

But the shape is too complex, do I really need to do the $P=12$ degree polynomial???

Nope, combine previous two approaches:

1) Divide interval into several parts

2) Fit simple, say 3-degree polynomial in it

But what if I need to have a continuous function,
not some unrelated polynomial parts?

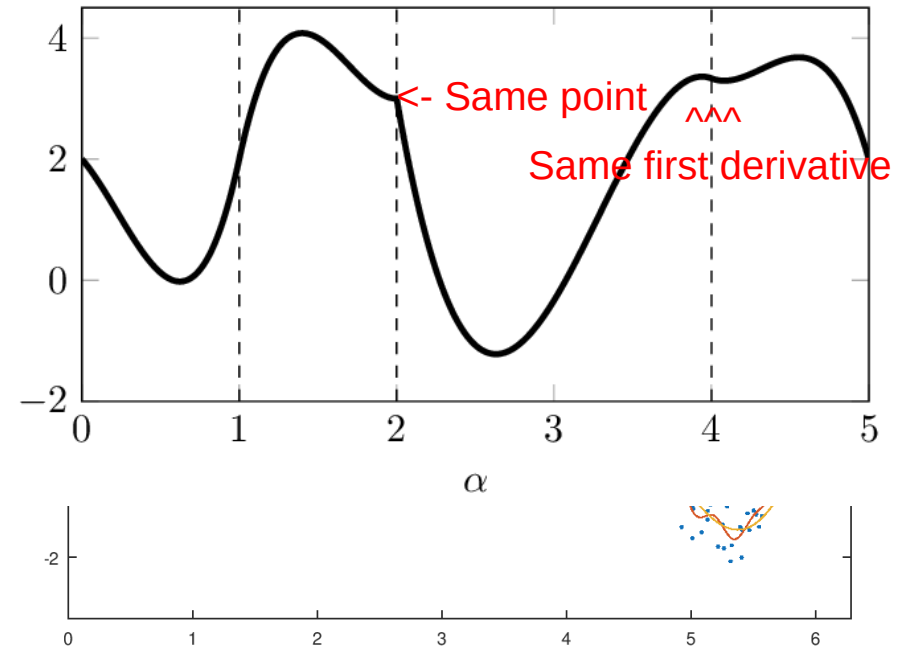
E.g.:

$$Y_1 = \beta_{10} + \beta_{11} \cdot X + \beta_{12} \cdot X^2 + \dots + \beta_{1P} \cdot X^P + \epsilon, \quad \text{if } X < C \quad A(\alpha)$$

$$\epsilon \in N(0, \sigma^2)$$

$$Y_2 = \beta_{20} + \beta_{21} \cdot X + \beta_{22} \cdot X^2 + \dots + \beta_{2P} \cdot X^P + \epsilon, \quad \text{if } X \geq C$$

$$\epsilon \in N(0, \sigma^2)$$



+ Continuity constraints:

$$Y_1(C) = Y_2(C) \quad (\text{same point, 0-continuity})$$

$$Y_1'(C) = Y_2'(C) \quad (\text{same first derivative, 1-continuity})$$

$$Y_1''(C) = Y_2''(C) \quad (\text{same second derivative, 2-continuity})$$

TOO HARD!

Splines will ensure maximum continuity (d-1 for polynom with degree d) with an easy formula:

- we can represent this model with truncated power basis functions

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_{K+3} b_{K+3}(x_i) + \epsilon_i$$

Splines

- where the b_k are **basis functions**

$$b_1(x_i) = x_i <$$

$$b_2(x_i) = x_i^2 < \text{construct first interval polynom}$$

$$b_3(x_i) = x_i^3 <$$

$$b_{k+3}(x_i) = (x_i - \xi_k)_+^3 \quad k = 1, \dots, K < \text{Every time we hit another knot,}$$

We change all coefficients (beta0 – beta 3).

Largest degree (3 for cubic) encapsulates every other term inside:

$$x^3 - A x^2 + B x - C$$

- where

$$(x_i - \xi_k)_+^3 = \begin{cases} (x_i - \xi_k)^3 & \text{if } x_i > \xi_k \\ 0 & \text{otherwise} \end{cases}$$

We don't have time to cover the math, refer to the lectures!

We don't have time to cover the smoothing splines either!

Okay, I have a [Polynomial/Step function/Piece-wise poly/Spline/.....],

but how do I know which

[degree of polynom/ number of interval points(knots) / .../ any other non-LinearReg parameter]

To choose?

Choose a set of models: $degree = 4, degree = 5, \dots, degree = 12$
 $number\ of\ intervals: 4, 5, 6, \dots, 11$

1) Split data into 10 folds

$model_1 = [degree = 4, knots = 4],$

2) For every model:

$model_2 = [degree = 4, knots = 5], \dots$

Cross validation!

For every possible fold id(1, 2, ..., 10):

2.0) Combine 9/10 folds into train, 1/10 into test set

2.1) fit on 9/10 train folds

2.2) Compute RSS/F-stat/adjusted R^2 on 1/10 test fold

3) Average test errors from 2.2) for across all fold id

4) Choose the best model by averaged score from 3)

Or you could proceed with ANOVA/F-test comparison or other way of comparing nested models

$$Y = \beta_0 + \beta_1 \cdot X + \epsilon,$$
$$\epsilon \in N(0, \sigma^2)$$

$$Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot X_2 + \epsilon,$$
$$\epsilon \in N(0, \sigma^2)$$

=====

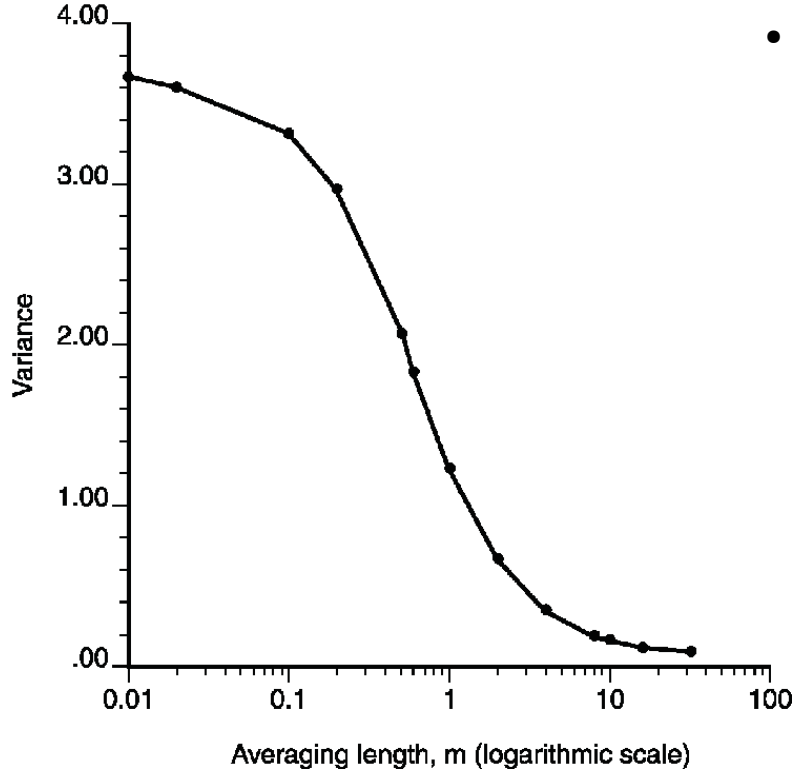
$$Y = \beta_0 + \beta_1 \cdot X + \beta_2 \cdot X^2 + \epsilon,$$
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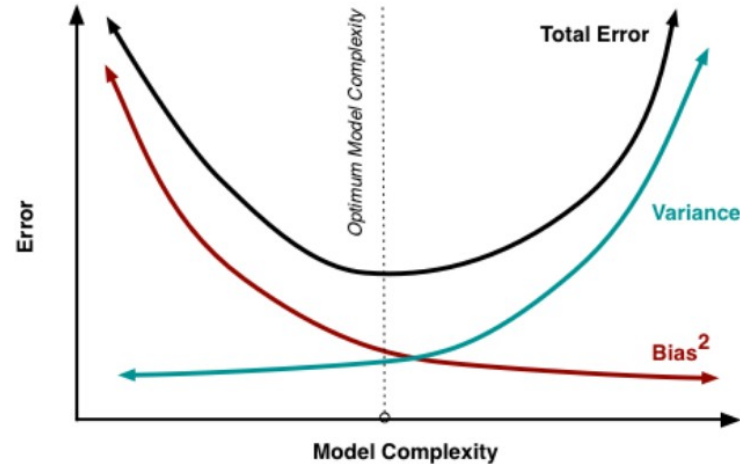
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$$Y = \beta_0 + \beta_1 \cdot X + \beta_2 \cdot X^2 + \beta_3 \cdot X^3 + \epsilon,$$
$$\epsilon \in N(0, \sigma^2)$$

And then just use **AIC** / **F-stat as in ANOVA** / ~~adjusted R²~~ (cannot use for non linear models!!!!
Not measuring quality anymore)



- You can think of this as following:
 - Addition of more variables (**or degrees in polynomial**) makes the prediction more precise (smaller variance)
 - But decrease from 1.1% to 1.09% is too small and the computational complexity is spent for nothing
 - Moreover, with more variables we need more and more samples to maintain same variance quality
 - Bias-variance tradeoff
 - **Is exactly how Cross-validation works!**

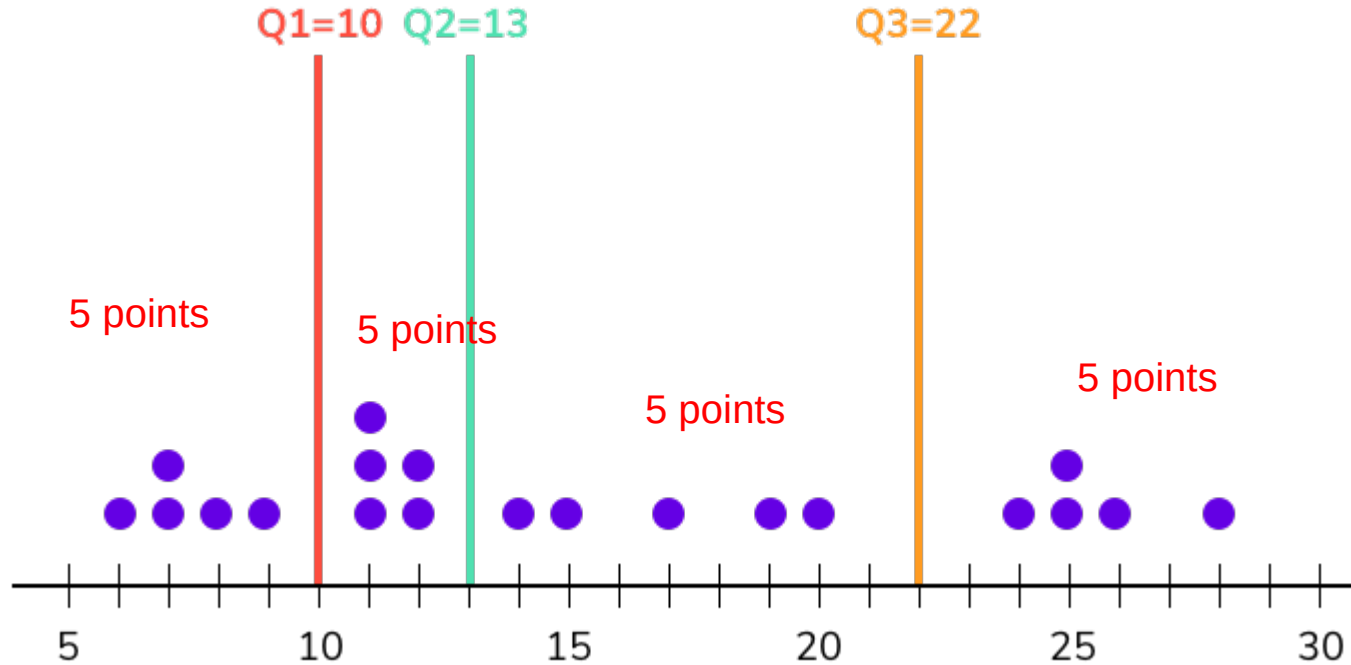


We can choose the position of knots if number of them is given:

quantiles =

such that each interval between knots has the same number of points

> optimality control via Cross-Validation

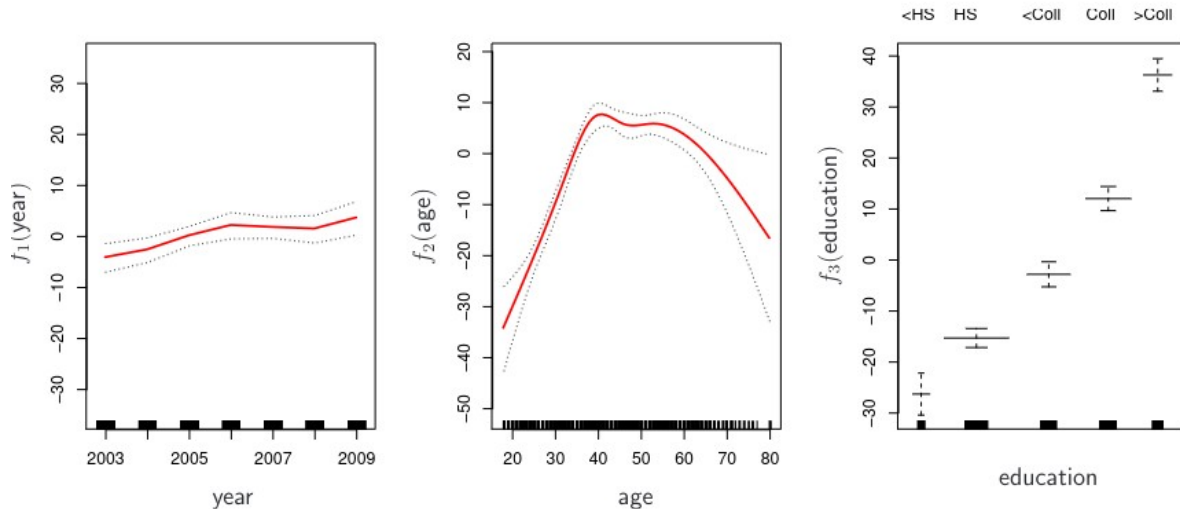


Okay, finally I can do it for 1 variable, but what if two variables have different models? How to combine them?

> Just add them and assume additivity(no interactions)

$$y_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}) + \epsilon_i$$

> Or, if interaction are REALLY needed,
R library will do it for you!

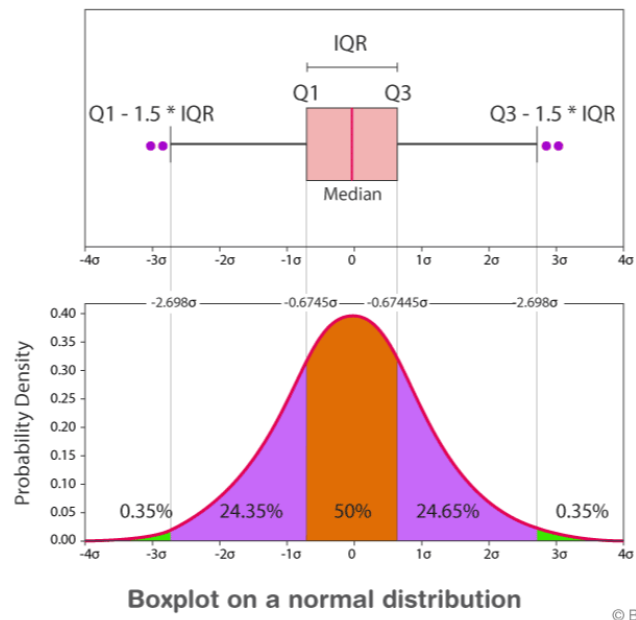


`gam(... + ns(age,df=5):ns(year,df=5))`

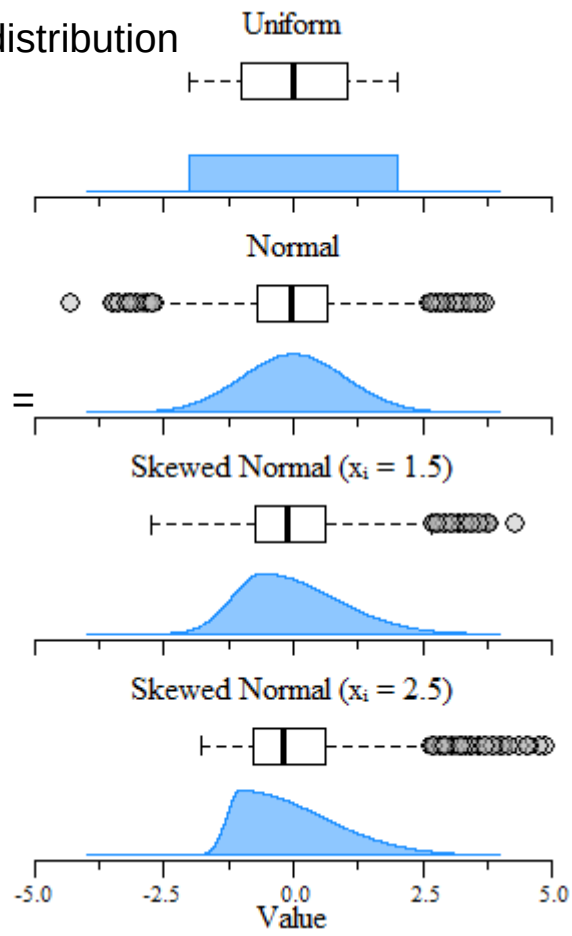
```
gam(  
wage ~ ns(year, df = 5) + ns(age, df = 5) + education  
)
```

Generalized Additive Models

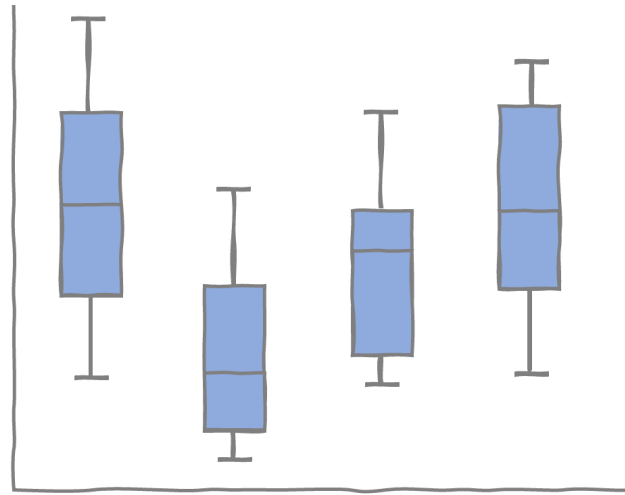
Some strange plot? Boxploty = categorical X distribution



X 3 groups =



Remove distribution plots



Categorical

Nominal

Ordinal



Pen



Pencil



Eraser



Cow



Dog



Cat



Excellent



Good



Bad



Fantastic

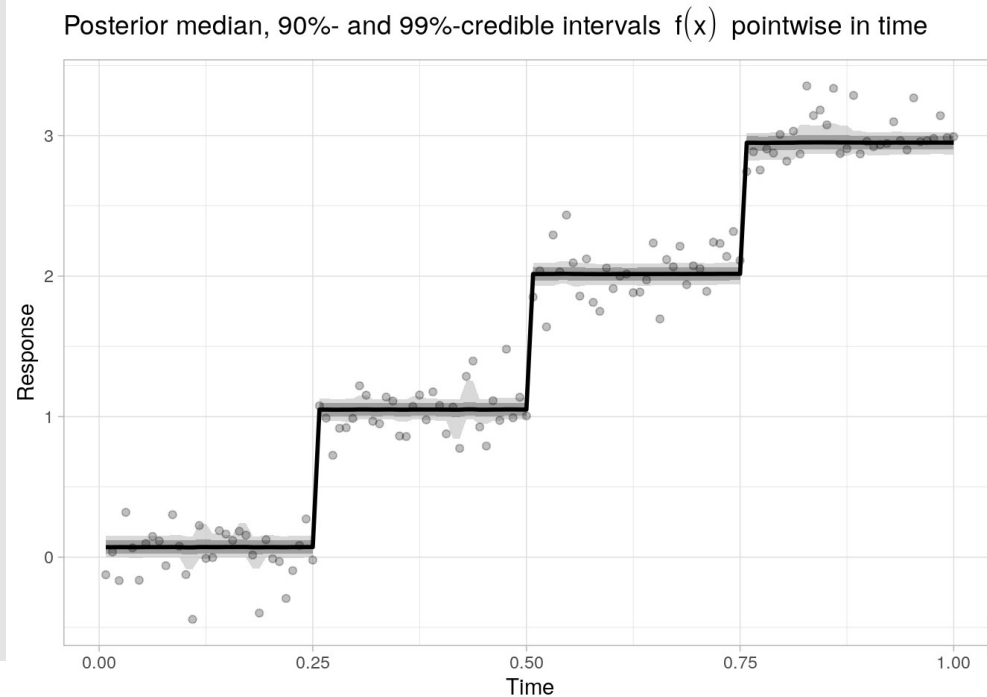
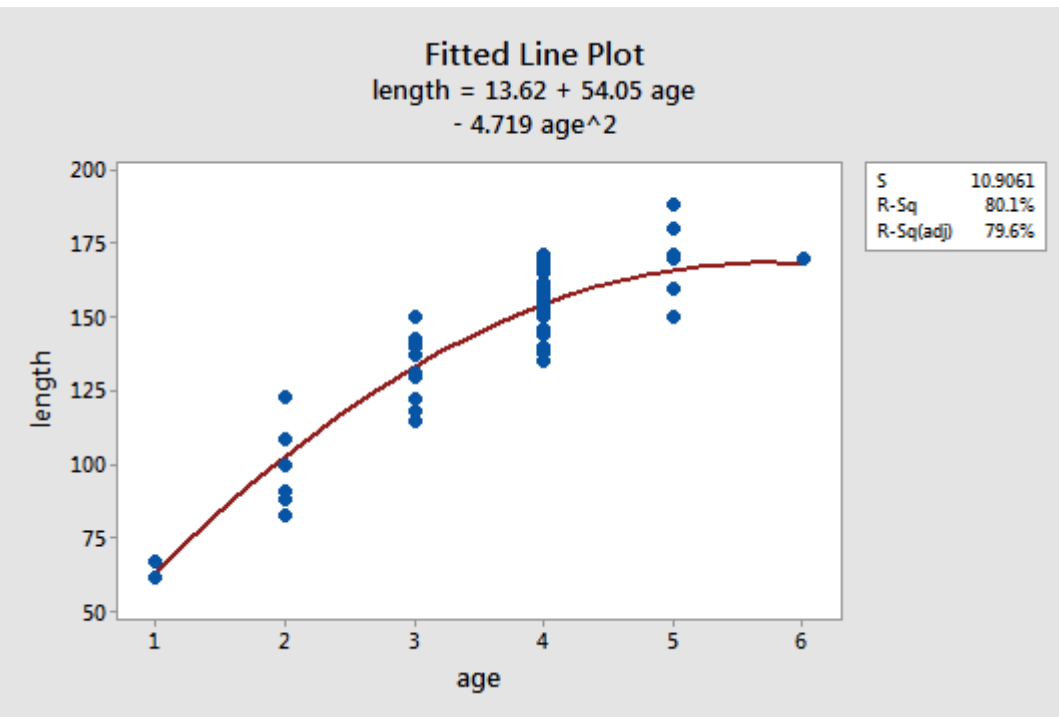


Okay



Don't Like

Categorical(Discrete) X? Just use polynomial or step function



Seminar 4: Nonlinear_wage.Rmd / html

Further questions to answer (homework):

1. Have a look at other predictors. What treatment would you recommend for them?
2. Which non-linear model would you recommend for wage prediction (considering all the predictors)? Show a model that improves the gam model tested in the last chunk.

Small help:

- Choose 3-5 predictor X variables
- For each/combo of them perform the analysis of assumptions (Qqplot, etc)
 - try to derive a type for each variable (polynomial? Step?)
 - Try to check
- Combine them into a single GAM and try to decide which additive parts of GAM are to be removed

